

R491: Outer Bridges III

TFPT RH programme

31 August 2026

Verdict: HIDDEN GAP FOUND / DENSITY BRIDGE REDUCED. The requested $8 \rightarrow 7$ census reduction is not attained. This note records the corrected target and does not make an RH claim.

1 Why the advertised two-piece closure is not valid

The r489 Lean definition is

$$\begin{aligned} & \text{FullWeilDyadicSampleConvergence} \\ & \quad \wedge \text{FullWeilArchContinuity} \\ & \quad \wedge \text{GridPoleHatIntegralIdentity}. \end{aligned}$$

Thus the completion has three components, not two. In particular the arch component cannot be omitted. Before r491, `fullWeilArchSide` was opaque, so no theorem could derive its limit from the topology regardless of the quality of the grid approximation. R491 replaces it by the concrete r475 regularized u -space pairing.

There were two further under-specifications. The L^2 witness had no relation between its support diameter and `supportRadius`; and uniform plus L^1 convergence alone does not control a weight asymptotic to $1/u$ at zero. The witness now has a support interval of length R and a Lipschitz constant.

2 Polar sign repair

The r376 potential and pairing are

$$\Pi(t) = -8(\cosh(|t|/2) - 1), \quad P_D(g) = - \sum_k g(kD) \frac{\Delta_D^2 \Pi(kD)}{D}.$$

Since $\Pi''(t) = -2 \cosh(t/2)$, the outer minus gives

$$P(g) = \int_{\mathbb{R}} 2 \cosh(u/2) g(u) du,$$

not the negative weight written in r489. For the one-cell test $D = 1, x_0 = 1$, both the discrete pairing and the positive integral equal 2.04201544330209; the negative integral has the opposite sign. Lean's `fullWeilPoleWeight` is corrected accordingly.

3 Explicit dyadic construction

R491 defines the represented half-open step function `GridElement.toStepFun` and the actual sampled sequence

$$D_m = 2^{-m}, \quad N_m = \lfloor R2^m \rfloor, \quad x_i^{(m)} = h(a + iD_m).$$

This is `dyadicSampleGridharm`. The values are arbitrary reals: `GridElement.x:Finsteps->Real` has no value quantization. Lean proves

$$N_m D_m \leq R$$

as `dyadicSampleGrid_supportBound_le`.

The exact remaining convergence target is `FullWeilDyadicSampleConvergence`. It asks this explicit sequence to satisfy the r489 fixed-support uniform-plus- L^1 topology. Its implication to the former existential density proposition is proved.

Mathlib’s `MemLp.exists_simpleFunc_eLpNorm_sub_lt` approximates by general measurable simple functions. It does not specialize their level sets to dyadic intervals. The explicit route must therefore prove the sampled-step L^2 estimate from the Lipschitz/support contracts and then the autocorrelation estimate

$$\|h_m \star \tilde{h}_m - h \star \tilde{h}\|_\infty \leq (\|h_m\|_2 + \|h\|_2) \|h_m - h\|_2.$$

4 Exact pole and arch remainders

The pole seam is narrowed to `GridPoleHatIntegralIdentity`. It says that every finite piecewise-linear ACF equals its hat expansion and that integrating each hat against $-\Pi'' = 2 \cosh(u/2)$ produces the second difference in `polePairingZ`. The implication from this pointwise identity to the sequence-level pole dictionary is proved.

For the arch channel, a valid proof needs a common Dini/Lipschitz majorant near $u = 0$, plus invariance of the regularized formula when the integration endpoint is enlarged beyond the actual grid support. Uniform plus L^1 convergence by itself is insufficient for the singular kernel.

5 Census and remaining outer bridges

The direct Lean census remains eight, with three sorries in `ExternalBridges.lean`: the corrected completion package, standard explicit-formula identification, and off-critical separation. Hence the density bridge is reduced but not closed.

After its eventual closure, bridge 2 still needs the full Guinand–Weil arithmetic/spectral dictionary and zero-multiplicity infrastructure absent from Mathlib. Bridge 3 still needs the off-critical separating smooth autocorrelation; the logical conversion to Mathlib’s `RiemannHypothesis` is already proved.